

Sleep and Memory Report (APA 7 Structured Sections)

Title Page

Sleep, Cueing, and Recall Performance in a 2x2 Between-Participants Study

Introduction

Sleep is often associated with memory consolidation, while targeted memory reactivation (TMR) is hypothesized to improve recall beyond control conditions.

Study Summary

Dataset size: N = 80 participants.

Design: 2x2 between-participants factors: Sleep (Sleep vs Wake) and Cue (TMR vs Control).

Outcome: recall_score (0 to 40).

Method

Participants completed a word-pair learning and recall task under assigned sleep and cue conditions. This report contains descriptive statistics only; no inferential claims are made.

Results: Overall Descriptive Statistics

Overall recall score: M = 25.62, SD = 4.60, Median = 25.19, Range = [14.43, 35.64]

Results: By Sleep Condition

Sleep: n = 40, M = 28.17, SD = 4.07, Median = 28.17, Range = [21.22, 35.64]
Wake: n = 40, M = 23.06, SD = 3.58, Median = 23.50, Range = [14.43, 30.73]

Results: By Cue Condition

TMR: n = 40, M = 27.19, SD = 4.45, Median = 26.64, Range = [17.44, 35.07]
Control: n = 40, M = 24.04, SD = 4.23, Median = 24.20, Range = [14.43, 35.64]

Results: Cell-Level Statistics (Sleep x Cue)

Sleep-Control: n = 20, M = 26.49, SD = 3.58
Sleep-TMR: n = 20, M = 29.85, SD = 3.92
Wake-Control: n = 20, M = 21.58, SD = 3.36
Wake-TMR: n = 20, M = 24.54, SD = 3.22

Complete Provided Data (All Rows)

id | sleep | cue | recall_score

1 | Sleep | TMR | 31.37
2 | Sleep | TMR | 25.32
3 | Sleep | TMR | 33.38
4 | Sleep | TMR | 34.23
5 | Sleep | TMR | 21.22
6 | Sleep | TMR | 24.14
7 | Sleep | TMR | 30.58

8 | Sleep | TMR | 28.58
9 | Sleep | TMR | 29.92
10 | Sleep | TMR | 26.16
11 | Sleep | TMR | 33.96
12 | Sleep | TMR | 33.5
13 | Sleep | TMR | 30.3
14 | Sleep | TMR | 35.07
15 | Sleep | TMR | 32.1
16 | Sleep | TMR | 26.13
17 | Sleep | TMR | 31.66
18 | Sleep | TMR | 25.69
19 | Sleep | TMR | 33.95
20 | Sleep | TMR | 29.78
21 | Sleep | Control | 25.17
22 | Sleep | Control | 22.94
23 | Sleep | Control | 31.5
24 | Sleep | Control | 25.3
25 | Sleep | Control | 24.07
26 | Sleep | Control | 24.42
27 | Sleep | Control | 28.4
28 | Sleep | Control | 27.64
29 | Sleep | Control | 27.86
30 | Sleep | Control | 27.94
31 | Sleep | Control | 35.64
32 | Sleep | Control | 24.17
33 | Sleep | Control | 23.69
34 | Sleep | Control | 22.34
35 | Sleep | Control | 28.77
36 | Sleep | Control | 31.08
37 | Sleep | Control | 25.49
38 | Sleep | Control | 22.22
39 | Sleep | Control | 22.29
40 | Sleep | Control | 28.93
41 | Wake | TMR | 27.34
42 | Wake | TMR | 26.44
43 | Wake | TMR | 21.01
44 | Wake | TMR | 25.04
45 | Wake | TMR | 24.53
46 | Wake | TMR | 24.98
47 | Wake | TMR | 27.92
48 | Wake | TMR | 25.01
49 | Wake | TMR | 27.06
50 | Wake | TMR | 24.3
51 | Wake | TMR | 25.3
52 | Wake | TMR | 26.84
53 | Wake | TMR | 17.44
54 | Wake | TMR | 22.56
55 | Wake | TMR | 21.88
56 | Wake | TMR | 21.13
57 | Wake | TMR | 22.76

58		Wake		TMR		30.73
59		Wake		TMR		20.1
60		Wake		TMR		28.36
61		Wake		Control		14.43
62		Wake		Control		20.49
63		Wake		Control		22.73
64		Wake		Control		24.64
65		Wake		Control		25.2
66		Wake		Control		25.57
67		Wake		Control		20.43
68		Wake		Control		19.92
69		Wake		Control		25.86
70		Wake		Control		21.14
71		Wake		Control		16.26
72		Wake		Control		16.9
73		Wake		Control		17.86
74		Wake		Control		24.24
75		Wake		Control		22.64
76		Wake		Control		25.11
77		Wake		Control		20.08
78		Wake		Control		22.71
79		Wake		Control		24.82
80		Wake		Control		20.61

Interactive Graph

An interactive Plotly graph is available at: reports/sleep_memory_interactive_graph.html

Data Integrity Note

All values in this PDF were computed or transcribed from the provided CSV without fabrication.